

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 30682

LM RELATIVISTIC ASTROPHYSICS

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

ANY CALCULATOR

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. Consider a neutron star-black hole binary system in circular orbit. The black hole has a mass $M_{\text{BH}} = 4 M_{\odot}$ and the neutron star a mass $m_{\text{ns}} = 1.7 M_{\odot}$.

The neutron star has a radius $R_{\text{ns}} = 12 \text{ km}$, and spins at a tenth of its break-up frequency.

The black hole has a spin $a_{\text{BH}}/M_{\text{BH}} = 0.4$.

- (a) Compute the rotation period of the neutron star. [4]

- (b) Compute the dimensionless spin of the neutron star, $a_{\text{ns}}/m_{\text{ns}}$. State clearly your assumptions. [5]

- (c) When the gravitational-wave frequency is 10 Hz, the black hole spin is aligned along the direction of the orbital angular momentum and the neutron star spin is anti-aligned with respect to the orbital angular momentum. Compute the total angular momentum of the system. [7]

- (d) The neutron star-black hole orbital separation decreases due to gravitational radiation reaction. The neutron star is eventually tidally disrupted by the black hole. Compute the tidal disruption radius, *i.e.* the separation at which the neutron star is tidally disrupted. Discuss whether or not photons produced during the disruption can escape to infinity. [6]

- (e) A binary system of individual mass components $m_{1,2}$, total mass $M = m_1 + m_2$, reduced mass $\mu = m_1 m_2 / M$ and chirp mass $\mathcal{M} = (\mu/M)^{3/5} M$ loses energy through gravitational wave emission at a rate (in geometric units, $c = G = 1$)

$$\frac{dE}{dt} = -\frac{32}{5} (\pi f \mathcal{M})^{10/3},$$

which causes the gravitational-wave frequency, f , to change according to

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \mathcal{M}^{5/3} f^{11/3}.$$

ANY CALCULATOR

Compute the total energy that is emitted in gravitational waves during the entire lifetime of this binary. Express the result as a function of the reduced mass, μ . State clearly your assumptions.

[8]

2. The area of the horizon of a Kerr black hole of mass M and spin parameter a is

$$A = 8\pi M \left[M + (M^2 - a^2)^{1/2} \right] .$$

- (a) Use Hawking's area theorem to find the minimum mass of a Schwarzschild black hole that results from the collision of two Kerr black holes of equal mass m and opposite angular momentum parameter a . **[8]**
- (b) Compute in the limit $|a| \rightarrow m$ how much energy can be radiated away. Express the result as a function of m . **[6]**
- (c) Compute a limit on the energy that can be radiated away if $a = 0$ as a function of m . **[6]**
- (d) Assuming that the energy is released on the dynamical timescale of the merger of these two black holes, provide an order of magnitude estimate of the luminosity produced during the merger. **[4]**
- (e) If the two black holes have each a mass of $10^7 M_\odot$, estimate the fraction of the total radiated energy that needs to be converted into electro-magnetic radiation during the collision to produce a luminosity comparable to the Eddington limit, corresponding to

$$L_{\text{edd}} = 1.3 \times 10^{38} \left(\frac{M}{M_\odot} \right) \text{ erg s}^{-1} ,$$

- and estimate how long this transient would last. **[6]**

3. Highly energetic protons in the interstellar medium interact with the photons of the cosmic microwave background, and some of them eventually enter the Earth's atmosphere.

(a) A highly energetic proton enters the Earth's atmosphere and produces a shower of particles whose total energy is 3×10^{20} eV. Compute $1 - v/c$, the fractional difference of the speed of the proton, v , with respect to the speed of light. The proton's mass is 938.3 MeV. [5]

(b) In its journey to the Earth, the proton traveled through the photons of the cosmic microwave background.

i. Write an expression for the proton's mean-free-path as a function of the photon number density, n_γ , and the proton-photon interaction cross-section, σ . [2]

ii. By estimating the proton's mean-free-path, using $n_\gamma \approx 400 \text{ cm}^{-3}$ and $\sigma \approx 2 \times 10^{-28} \text{ cm}^2$, discuss whether the source of this high energy proton could be a physical process at high redshift. [2]

(c) The four-momentum of a particle of mass m , energy E and three-momentum $\vec{p}$ is $\mathbf{p} = (E, \vec{p})$. The inner-product between two four-vectors, $\mathbf{a}$ and $\mathbf{b}$, as a function of their components is: $\mathbf{a} \cdot \mathbf{b} = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3$.

Consider the collision of a proton with a cosmic microwave background photon leading to the production of a neutron and a pion. The energy in the interaction is at threshold, and the resulting neutron and pion of mass m_n and m_π , respectively, are at rest in the centre of mass frame.

i. What is the total three-momentum, $\vec{p}_n + \vec{p}_\pi$, of the neutron and pion in the centre of mass frame? [3]

ii. What is the magnitude of the total four-momentum $(\mathbf{p}_n + \mathbf{p}_\pi)^2$ in the centre of mass frame? Express your result as a function of m_n and m_π . [5]

ANY CALCULATOR

In the cosmic-microwave-background frame, assume that the proton moves along the x -axis with energy $E_p \gg m_p$ and the collision is with a photon of energy E_γ travelling in the opposite direction.

- iii. Write the components of the four-momentum of the proton and photon, $\mathbf{p}_p$ and $\mathbf{p}_\gamma$, respectively, as a function of E_p and E_γ . **[6]**

- iv. The total four-momentum in the interaction is conserved. Hence, derive an expression for E_p as function of m_p , m_π and E_γ . You can assume $m_p \approx m_n$. **[6]**

- v. Compute the proton's energy E_p necessary for a neutron and pion to be produced. Cosmic-microwave-background photons have typical energies $E_\gamma \approx 6 \times 10^{-4}$ eV, and the mass of a pion is $m_\pi = 139.6$ MeV. **[1]**

Formula Sheet

Required module specific formulae are given in the exam questions themselves.

Astrophysical Constants and Units

Astronomical Unit	AU	$1.50 \times 10^{11} \text{ m}$
Parsec	pc	$3.1 \times 10^{16} \text{ m}$
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_{\odot}$	$3.84 \times 10^{26} \text{ W}$
Absolute bolometric magnitude of the Sun	M_{bol}	$+4^{\text{m}}75$
Bolometric correction for the Sun	BC	$-0^{\text{m}}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\text{m}}74$
Solar constant	f	$1.36 \times 10^3 \text{ W m}^{-2}$
Solar mass	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Wien constant	b	$2.898 \times 10^{-3} \text{ m K}$
Hubble constant	H_o	$70 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Solar radius	$R_{\odot}$	$6.95 \times 10^8 \text{ m}$
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_{\oplus}$	$5.972 \times 10^{24} \text{ kg}$
Earth radius	$R_{\oplus}$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.